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Air diffusion performance index (ADPI) of overhead-air-distribution at low cooling loads

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ABSTRACT

The air diffusion performance index (ADPI) concept and its implementation in the diffuser selection guideline (*ASHRAE Handbook-Fundamentals, 2009*) provide designers a simple method to design overhead-air-distribution systems. However, the current guideline considers only five diffuser types and the range of sensible cooling loads used to develop the guideline (65 W/m^2–250 W/m^2) is high for current buildings. Applying this outdated guideline may cause draft discomfort and/or unnecessary fan-energy waste. This paper presents new experimental results to update the ADPI-based guideline to include 15 diffuser types at low cooling loads of 25 W/m^2 and 50 W/m^2. Compared to the high loads in the current guideline, the ADPI values generated in this work are greater due to the lower loads used. Most ceiling-mounted diffusers produce higher ADPI at a large range of airflow than do sidewall and other diffuser types. ADPI for ceiling diffusers has the least variation with changes in loads or airflow rates. The results show that the minimum airflow set-point for a variable air volume system varies with diffuser type. For ceiling diffusers other than round and perforated diffusers, the minimum set-point can be reduced to lower than 30% of the peak flow rate without significantly reducing ADPI.

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1. Introduction

Traditional overhead air distribution systems are designed to create sufficient air mixing and to avoid discomfort due to draft in the occupied zone. One essential procedure in heating, ventilating and air-conditioning (HVAC) design involves selecting and positioning terminal diffusers based on an associated air diffusion performance index (ADPI) specified in the ASHRAE handbooks and standards [1–4]. ADPI, reported as a number between zero and one, is meant to be the percentage of a space meeting the criteria of local effective draft temperature and air speed, which is defined differently for cooling and heating modes [1,5]. ADPI reflects room air uniformity and includes some modifications for draft [6–9]. The index also predicts indoor air distribution created by HVAC diffusers at a certain airflow rate and room thermal load. As indoor air distribution in a room depends on many factors, including diffuser geometry and installation configuration, airflow rate, room load, and room size [10,11], the ADPI method provides a simple tool to evaluate the complex process a designer undertakes to ensure air distribution in a space, which includes positioning diffusers, selecting diffuser types, and assessing air temperature and speed distributions.

ADPI has been adopted in ASHRAE handbooks and used in HVAC design since the 1970s [12]. Miller [11] introduced the concept of ADPI in an ASHRAE research project and measured ADPI for four types of outlets under different loads and airflow rates. However, until now the ADPI guideline in ASHRAE included only five diffuser types (Table 1) at cooling loads between 65 W/m^2 and 250 W/m^2–much higher than today's values [2,3]. Driven by concerns for building energy-efficiency, both new and retrofitted buildings have reduced room cooling loads through various measures, such as using higher insulation materials and energy-efficient lighting systems. Schiavon et al. [13] have shown that indoor cooling load varies from 20 W/m^2 to 172 W/m^2, with a median equal to 51 W/m^2. Arens et al. [14] observed that the most frequently observed loads were around 15–20 W/m^2 during the hottest month of California in two Yahoo! buildings. These values are obviously much lower than the loads prevalent at the time when the current ADPI guideline was developed. Therefore, the guideline needs to be updated with loads more applicable to current building design.

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Table 1
Diffuser types listed in current ASHRAE handbooks [2,3].

Diffuser type	Installation location
Grills	High sidewall
Grilles (Spread Vanes)	Window sill
Grilles (Straight Vanes)	Window sill
Circular Ceiling Diffusers	Ceiling
Ceiling Slot Diffusers	Ceiling
Light Troffer Diffusers	Ceiling
Cross-Flow Pattern Diffusers	Ceiling

The updating of ADPI for low cooling loads has energy implications for both constant air volume (CAV) and variable air volume (VAV) systems.

A CAV system needs a cooler supply air when higher cooling loads are present. Since buoyancy causes cold air to fall directly downward into the occupied zone rather than spreading along the ceiling when the diffuser jet momentum is not strong enough, indoor air distribution with a higher cooling load typically has a lower ADPI [10]. At such conditions, diffusers often require more airflow to provide sufficient mixing and maintain a high ADPI in the room. The ASHRAE guideline stipulates an ADPI of 80% as the minimum acceptable level for cooling loads up to 250 W/m^2. Because today's cooling loads are much less than those specified by ASHRAE, however, design airflow rates and fan energy consumption in a building designed with the current guideline will be larger than necessary. Even worse, designers or practitioners might use an arbitrary airflow rate of their own choosing due to the lack of ADPI guidance for low cooling loads, causing issues of insufficient diffuser mixing and draft discomfort because of buoyancy.

A VAV system, on the other hand, adjusts airflow rate with the variation of room load while maintaining supply air temperature. VAV systems need to provide an ADPI greater than 80% at the full range of operating airflow. The minimum airflow set-point of a VAV system depends on the requirements for ventilation, thermal load removal and thermal comfort. The conventional minimum set-point is 30%–50% of the maximum flow [15–17]. However, a case study in an office found that 15% of the design airflow can still maintain control stability and comply with ASHRAE standards for ventilation and comfort [18]. Another field study sponsored by ASHRAE in seven buildings supported that finding and showed that the minimum set-point could be decreased to 10%–20% without reducing comfort or air quality significantly [14]. This field study also pointed out that two (round ceiling diffuser and high sidewall grille) of six tested diffuser types produced a considerable decrease in ADPI for low airflow rates that may often be related to low cooling loads. Those findings suggest the need for further research on ADPI for various diffuser types at low cooling loads.

The objective of the present study is to update and expand the ADPI guideline with more diffuser types and under lower cooling load conditions. This paper also shows how ADPI varies with airflow rate, which helps for predicting the performance of diffusers when implemented in VAV systems.

2. Air diffusion performance index

The air diffusion performance index (ADPI) concept combines local air speed and local dry-bulb temperature into an effective draft temperature (Eq. (1)) [1]. The effective draft temperature reflects the subjective sensation of coldness in terms of air motion and skin temperature deviation from surroundings [9]. ADPI is calculated as the percentage of an occupied zone within the ranges of acceptable air speeds ($\leq$0.35 m/s) and effective draft temperatures (−1.7 to 1.1 °C) [1].

$$\mathrm{EDT} = t_i - t_a - 8.0(V_i - 0.15)^\circ C \quad (1)$$

where t_i is air temperature at a test point, i; t_a is spatial average air temperature in the entire occupied zone and V_i is local air speed. The acceptable range of EDT is between −1.7 °C and 1.1 °C in cooling mode.

We measured indoor air temperatures and speeds (Eq. (1)) at four horizontal planes (0.1 m, 0.6 m, 1.1 m, and 1.7 m from the floor) with 15 evenly distributed locations in each plane. Previous literature reported more details of the measurement positions [5,10,19]. Isothermal throw lengths (terminal speed of 0.25 m/s) at various flow rates for each tested diffuser were provided by manufacturers, which were measured according to the test procedure in ASHRAE Standard 70 [4].

Miller and Nash [20] have shown that the ratio of the isothermal throw length (T) of a diffuser jet to the characteristic length (L) for a given room is a critical factor in determining ADPI in cooling mode. The ASHRAE handbooks [2,3] thus list ranges of T/L ratios within which ADPI is greater than or equal to 80% for different diffuser types. We used the same method in this study. For instance, the curve fitting in Fig. 1(a) shows that the acceptable T/L range (ADPI > 80%) is 1.7–2.2 for adjustable blade grilles. The range can be determined directly from measurements when the tested airflow rate per diffuser is between 230 m^3/h and 770 m^3/hr, because both the maximum and minimum airflow rates produce ADPI smaller than or approximately equal to 80%. However, ADPI for some other diffuser types, such as the linear slot diffuser in Fig. 1(b), remains greater than 80% for the whole test range of flow rates and T/L ratios. For this condition, we only report T/L ranges calculated from tested airflow rates rather than extrapolating to T/L values yielding ADPI smaller than 80%. The extrapolation in Fig. 1(b) shows that ADPI less than 80% occurs at T/L less than approximately 0.3 or greater than 4.8, corresponding to room air change ratesof 0.9 h^{-1} or 19 h^{-1}, respectively. Since the small air change rate requires an unrealistically low supply temperature and the high flow rate causes unacceptable pressure resistance and noise, we limited total tested flow rates in the room to between 230 m^3/h and 770 m^3/h for all diffusers, as described in section 3.1.

3. Methodology

In this study, we measured the air diffusion performance index (ADPI) of diffusers at low cooling loads (50 W/m^2 and 25 W/m^2) in a test room that resembled an office. We present here experimental setups and testing procedure. This section also describes the 15 types of overhead-air-distribution diffusers tested, which are commonly used in the market today.

3.1. Experimental setups and apparatus

A test chamber (Fig. 2a) with a dedicated HVAC system and state-of-the-art controls [21] was modified to house various types and sizes of diffusers. Below a suspended ceiling, the conditioned zone measured 5.2 m × 4.5 m × 2.4 m. Overhead air distribution was created by supplying air from one of four installation configurations: (1) ceiling center, (2) ceiling periphery, (3) high sidewall and (4) window sill. Fig. 2a illustrates the experimental setup for the four configurations. For the ceiling center configuration, we used two identical diffusers evenly spaced along the centerline of the long dimension of the room. For the other three configurations, we used only one diffuser in the center of the horizontal dimension of the shortest wall. The supply airflow rate in total ranged from 230 to 770 m^3/h. The range was selected by considering the minimum supply air temperature, ventilation requirement, and diffuser operating conditions such as noise criteria. The calculated air exchange rate (ACH) was between approximately 4 and 14 h^{-1}, which covers typical air change rates in offices and classrooms with mechani-

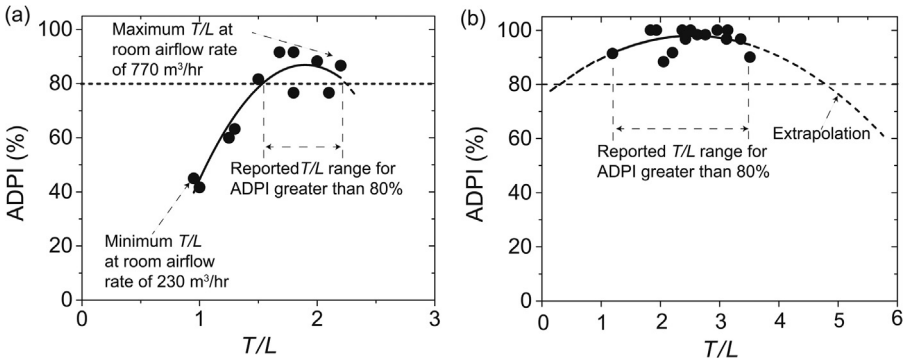


Fig. 1. Method to determine *T/L* ranges for ADPI greater than 80%. (a) Adjustable blade grills (horizontal blades) with ADPI greater than 80% at *T/L* of 1.7–2.2; The *T/L* range was determined based on 80% ADPI; (b) Linear slot diffusers with ADPI greater than 80% at all tested airflow rates; The *T/L* range was calculated based on total room airflow rate of 230 m^3/h and 770 m^3/h. The experiment used one adjustable blade grille for case (a) and two identical linear slot diffusers for case (b) in the tests. ADPI data was extracted from a previous study at a cooling load of 50 W/m^2 [10]. *T* is the throw length with a terminal speed of 0.25 m/s; *L* is room characteristic length.

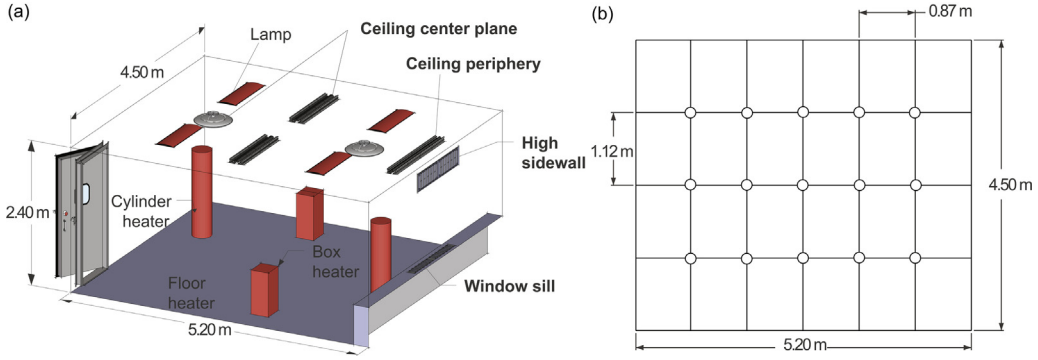


Fig. 2. Experimental setup for all four configurations. (a) Schematic of the experimental setups; (b) Measurement locations for air speed and temperature (15 positions for each horizontal plane: 0.1 m, 0.6 m, 1.1 m and 1.7 m above the floor).

Table 2
Cooling load simulators.

Heaters	Room Loads	
	25 W/m^2	50 W/m^2
Lamps	100 W × 4	100 W × 4
Box heaters	47 W × 2	100 W × 2
Cylinder heaters	47 W × 2	100 W × 2
Floor heater	0 W × 1	370 W × 1

cal HVAC systems [22–24]. The maximum temperature difference between supply and room air was 7 °C and 14 °C for cooling loads of 25 W/m^2 and 50 W/m^2, respectively.

A few power-adjustable electric heaters (Table 2) simulating occupants and office equipment provided indoor cooling loads. Distribution of these internal heat sources mimicked a typical office environment (Fig. 2a). In addition, the chamber floor was fully covered with an electric sheet-heater to model solar gains due to solar radiation absorbed by the floor. This study involved two cooling loads of 25 W/m^2 and 50 W/m^2 recommended by previous studies [13,14], and lower than those specified by ASHRAE [2,3]. Load variations were achieved by adjusting electric power for the heaters. Load control and energy balances were maintained by well-insulated wall surfaces that ensured heat losses through the envelope of less than 3% of the total.

We applied hot-sphere omni-directional-anemometers to measure spatial air temperature and speed at four horizontal planes (0.1 m, 0.6 m, 1.1 m, and 1.7 m from the floor) with 15 evenly distributed locations in each plane (Fig. 2b). The measured air speed and temperature was used to calculate ADPI, which is described in more detail in Section 2.

We also monitored air temperature and flow rate at supply and exhaust vents for each test. Table 3 summarizes the details of the used instruments in these experiments.

3.2. Studied diffuser types

After consulting with five diffuser manufacturers across the United States and Canada, we defined a list of the most commonly used diffuser types together with the project committee of ASHRAE RP-1546: Expansion and Updating of the Air Diffusion Performance Index method. The list is summarized in Table 4. For each diffuser type, we considered two sizes and two to four different manufacturers.

4. Results

We next present measured ADPI and *T/L* ranges for acceptable ADPI (>80%) for the 15 diffuser types at two cooling loads of 25 W/m^2 and 50 W/m^2.

Fig. 3 displays all ADPI data at various *T/L* ratios. Curve-fitting of measured data shows that ADPI depends primarily on *T/L* and secondarily on room load, especially for diffusers installed at the high sidewall configuration (Fig. 2). The general R^2 for regressions is 0.44 (0.27, 0.71) representing median (first, third quartile). Overall, ADPI decreases with increase in room cooling loads for almost all tested diffusers. However, ADPI varies the least for diffusers installed at the ceiling center plane, such as square ceiling diffusers, louvered face diffusers, plaque face diffusers, linear slot diffusers and swirl diffusers. The room load and flow rate have an insignificant effect on ADPI for the diffusers at the ceiling center, except for round ceiling diffusers and perforated diffusers. The findings suggest that

Table 3
A summary of the instruments and uncertainties.

Instrument	Model	Accuracy	Measured variables
Air flowmeter	EBTRON GT Type A 116	5% of reading	Supply and return airflow rate; Verified with a Pitot probe.
Electricity usage monitor	Kill-A-Watt, P4400	0.5% of reading	Heat-simulator power; The maximum accuracy of 2% of reading.
Omni-anemometer	HT-400, SENSOR, Poland	±0.03 m/s ±0.2 °C	Spatial air temperature and speed for ADPI calculation.
Thermistor	Model 44033, OMEGA	±0.1 °C	Supply and return air temperature.

Table 4
Tested diffuser types in this study.

Terminal Device	Size	Manufacturer[§]	Installation location
Adjustable Blade Grilles	41 × 15 & 61 × 15 cm	A, B & D	High sidewall (45° upward blades) High sidewall (0° horizontal blades) High sidewall (45° downward blades)
Fixed Blade Grilles	41 × 15 & 61 × 15 cm	A–D	High sidewall (15° upward blades) High sidewall (15° downward blades)
Linear Bar Grilles	41 × 15 & 61 × 15 cm	A–D	High sidewall Window sill
Nozzles	13[*] & 15[*] cm	A–C	High sidewall
Round Ceiling Diffusers	15[*] & 25[*] cm	A, B, D & E	Ceiling center plane
Square Ceiling Diffusers	15[*] & 20[*] cm	A–C	Ceiling center plane
Perforated Diffusers-Round Pattern	15[*] & 20[*] cm	A–C	Ceiling center plane
Perforated Diffusers-Directional Pattern (4-way)	15[*] & 20[*] cm	A–D	Ceiling center plane
Louvered face Diffusers- with Lip on Deflector Blade	15[*] & 23[*] cm	A–D	Ceiling center plane
Louvered face- without Lip on Deflector Blade	15[*] & 23[*] cm	D & E	Ceiling center plane
Plaque Face Diffusers	15[*] & 20[*] cm	A–D	Ceiling center plane
Linear Slot Diffusers	Two-way 2 & 4 slots[†], 120 cm long	A, B, D & E	Ceiling center plane
T-Bar Slot Diffusers	One-way 2 & 4 slots[†], 120 cm long	A, B, D & E	Ceiling periphery
Swirl Diffusers	15[*] & 20[*] cm	A–C	Ceiling center plane
N-slot Diffusers	1 slot[†], 120 cm long	A, C & D	Ceiling periphery

[§] A: Krueger; B: Price; C: Titus; D: Nailor; E: Metalaire.
[*] Size in duct diameter.
[†] Slot width: 1.9 cm.

those types of diffusers perform best for VAV systems. A high ADPI for those diffusers can be maintained even at a low airflow rate or high cooling load.

The minimum allowed flow rate is dependent on diffuser type. For most ceiling diffuser types, such as plaque face diffusers, high ADPI greater than 80% can be maintained at the entire range of tested flow rates from 230 to 770 m^3/h. The extrapolation of ADPI curves for plaque face diffusers implies that an airflow lower rate than 230 m^3/h (30% of the maximum flow rate of 770 m^3/h) likely generates ADPI greater than 80%. In other words, a lower airflow rate than the traditional minimum set-point (30%–50%) can still create comfort and sufficient mixing for ceiling diffusers except for round ceiling diffusers and perforated diffusers. However, diffusers installed on high sidewalls or other configurations may cause draft issues at low flow rates. A possible solution is to change the deflectors of those diffusers [10]. Fig. 3 shows that adjustable blade and fixed blade grilles with blades upwards perform much better than with other blade configurations.

To facilitate comparison among various diffuser types, we summarize *T/L* ranges for ADPI greater than or equal to 80% in Fig. 4. For some diffuser types, it is worth noting again that *T/L* ranges are limited to total room flow rates from 230 to 770 m^3/h (air exchange rate from 4 to 14 h^{-1}) rather than the cutoff of ADPI equal to 80%. The reported *T/L* ranges for ADPI > 80% might be extrapolated based on the regressed curves in Fig. 3. Table A1. in the Appendix A combines *T/L* ranges and regressed curves for each diffuser type.

ASHRAE [2,3] reports ranges of *T/L* for ADPI > 80% for the lowest cooling load of 33 W/m^2 for only high sidewall grilles and round (circular) ceiling diffusers, and 65 W/m^2 for the rest. Overall, the reported *T/L* ranges for almost all diffuser types at low cooling loads in this study are wider than what are specified in ASHRAE for high loads. For instance, the acceptable *T/L* range (ADPI > 80%) for sill grilles at 50 W/m^2 is 1.3–2.1 compared with 1.2–1.8 at 127 W/m^2.

The only exception is high sidewall grilles. The acceptable *T/L* is 0.7–2.1 for high sidewall grilles at <33 W/m^2, wider than what we measured (1.2–2.2) at 25 W/m^2. The issue happens for a higher cooling load as well: 1–1.9 at 65 W/m^2 in ASHRAE vs. 1.7–2.2 at 50 W/m^2 in this study. The major reason is that previous studies [11,25], from which the ASHRAE guideline was developed, located most heat sources at the bottom center of the wall where high sidewall grilles were installed. The buoyancy flow from the heat sources partially counteracted the low momentum of jets from the grilles, leading to a higher ADPI. The results imply that locations of heat sources could influence diffuser performance, because of altered air speeds by buoyancy [26].

Fig. 4 simplifies the acceptable *T/L* ranges for all 15 diffuser types. The figure provides a simple tool for HVAC engineers or designers to select proper diffusers. For the two cooling loads, the linear slot diffuser has the largest *T/L* ranges for ADPI greater than 80%, which would be suitable for the application of VAV systems.

5. Discussion

The results above have been used to update the ASHRAE ADPI guideline based on experiments in a chamber with fixed geometry. For each diffuser test, we installed one or two diffusers, as shown in Fig. 2a (two diffusers mounted at the ceiling center plane or just one at the high sidewall), in a room with a fixed height. However, questions might arise as to how this new design guideline can be applied to rooms with different heights or with different diffuser numbers and layouts. We discuss these concerns in the following section.

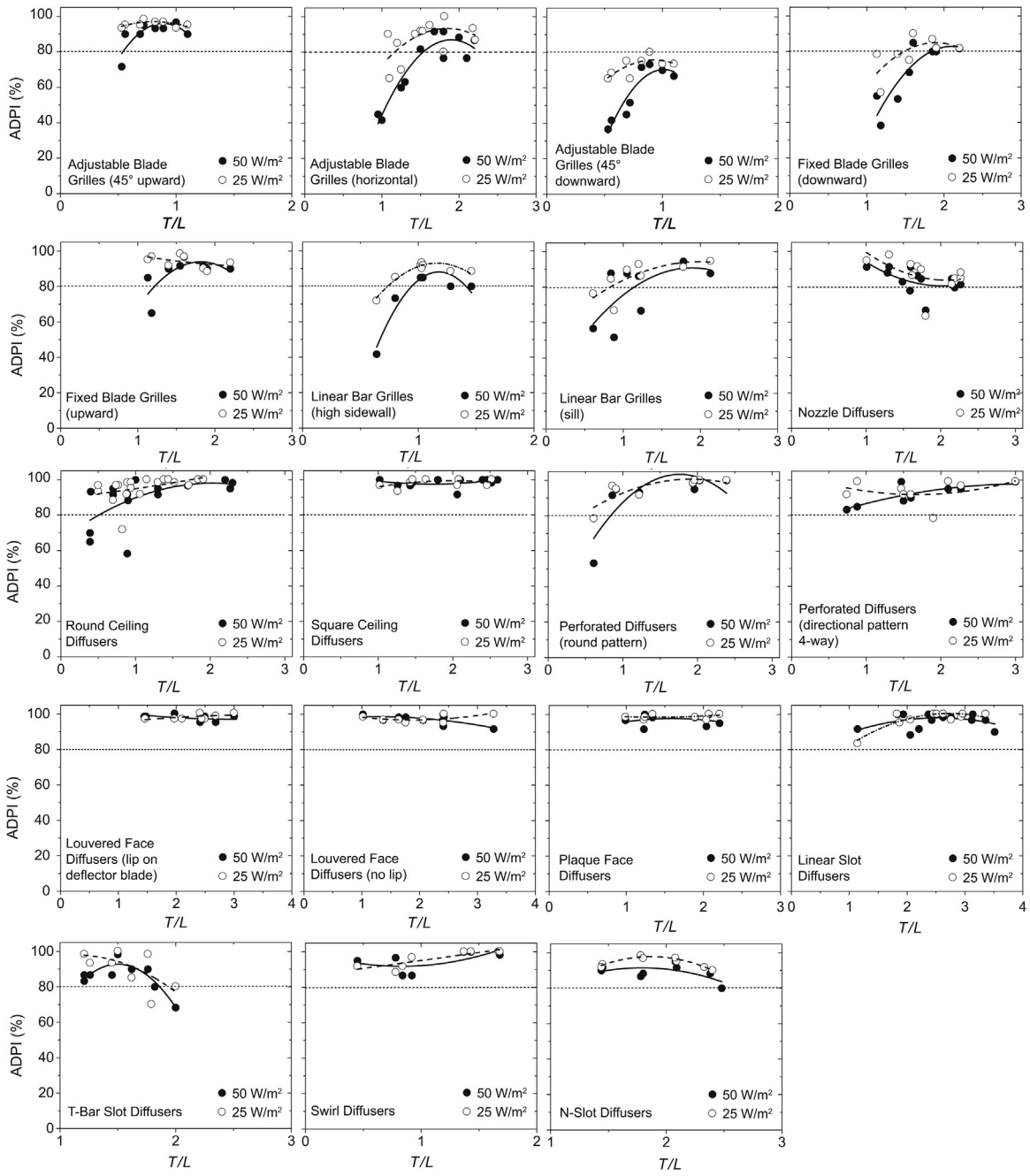


Fig. 3. ADPI for various diffuser types at low cooling loads. *T* is the throw length with a terminal speed of 0.25 m/s; *L* is room characteristic length.

5.1. The effect of room height

Our previous study [10] discussed how ADPI results developed from an office-like chamber may not be generalized to spaces with large volumes and high ceilings (e.g., airport terminals, auditoriums with sloped floors) in which diffuser jets are high above the occupied zone (<1.7 m). The normal height of an indoor space (e.g., office or classroom without a sloped floor) with all-air systems below the suspended ceiling is approximately 2.7 m. However, the test chamber has a height of only 2.4 m in this study. Typically, the space required for a diffuser jet to develop and to mix with surrounding air depends on diffuser features along with room and supply air conditions. ADPI will likely be increased with the presence of a larger mixing zone above the occupied zone.

Here we argue that the small difference in room height (2.4 m vs. 2.7 m) has an insignificant impact on ADPI results presented in this study. We conducted a sensitivity analysis to investigate to what extent ADPI results vary with space height above the occupied zone. Because we could not increase the test chamber height, we instead measured air speeds and temperatures in a subset of the occupied zone (1.35 m from the floor). We then compared the new ADPI, based on four new measurement heights (0.1 m, 0.6 m, 1.1 m and 1.35 m) with the results calculated with the standard (0.1 m, 0.6 m, 1.1 m and 1.7 m) specified in ASHRAE Standard 113 [1]. Ceiling diffusers typically have thinner jets than do high sidewall diffusers, which suggests that room height may have more influence on the performance of high sidewall grilles or nozzles. Therefore, we only considered the comparisons for adjustable blade grilles and nozzles

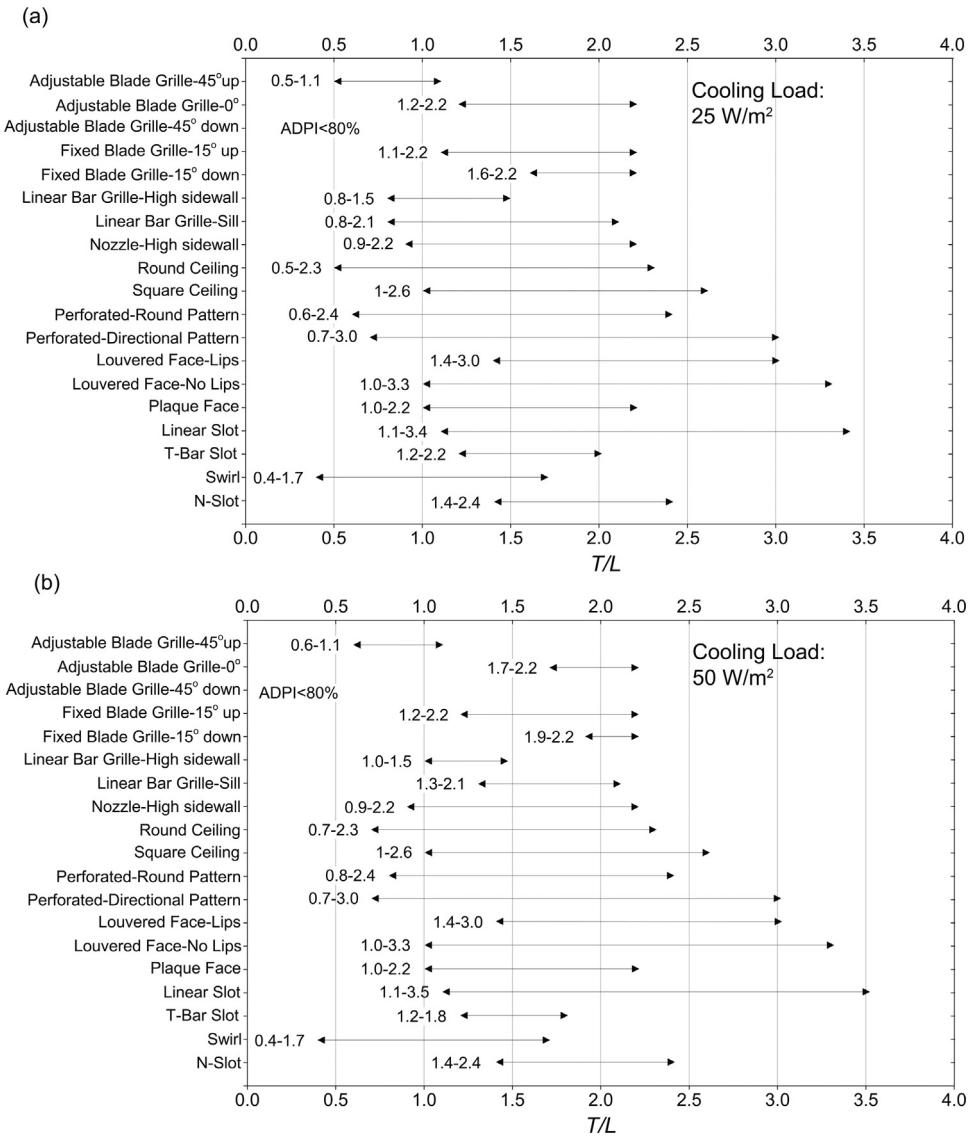


Fig. 4. *T/L* ranges for ADPI greater than 80% in cooling mode. (a) Cooling load of 25 W/m^2; (b) Cooling load of 50 W/m^2. *T* is the throw length with a terminal speed of 0.25 m/s; *L* is room characteristic length. The limits of a *T/L* range for a given diffuser type are based on ADPI equal to 80% or the room airflow rates between 230 m^3/h and 770 m^3/h.

of which the performances also vary widely with *T/L*, as suggested in Fig. 3.

Fig. 5 shows the comparisons of calculated ADPI using two different sets of sampling locations for adjustable blade grilles and nozzles at the cooling load of 50 W/m^2. ADPI was increased by less than 5% when a lower sample location was employed for both diffuser types. This finding indicates that a ceiling height of 2.4 m is sufficient for diffuser jets for mixing with surrounding air. It also implies that the ADPI results in this study can be applied for typical heights (e.g., 2.7 m) of indoor spaces.

5.2. The influence of floor area and number of diffusers

ADPI varies with the non-dimensionless ratio *T/L*. The throw length (T) is a function (Eq. (2)) of airflow rate (Q_o), terminal air speed (V_T) and diffuser features [2]. Eq. (2) shows that throw length of a diffuser jet varies proportionally with diffuser flow rate.

$$T = \frac{K_c Q_o}{V_T A_o^{0.5}} \quad (2)$$

where K_c is a centerline velocity constant; V_T is terminal air speed (0.25 m/s here), and A_o is diffuser free area.

A larger room typically has a greater characteristic length and requires more supply airflow. Given a fixed diffuser number and air exchange rate, supply airflow and throw length of each diffuser jet would be increased by the same factor as the increase in room size. Therefore, *T/L* and ADPI remain unchanged with the variation of floor area. In the Appendix A, we discuss briefly how diffuser number affects ADPI.

6. Conclusion

We updated and expanded, for the first time since the 1970s, the ASHRAE ADPI guideline for overhead air distribution systems. The new guideline considers 15 types of diffusers operating at lower cooling loads of 25 W/m^2 and 50 W/m^2, which covers the most used types and operating conditions in today's buildings.

In general, a lower cooling load produces a higher ADPI. The tested diffusers at low cooling loads can operate in wider ranges of airflow rates than what ASHRAE specifies for high cooling loads. The location of heat sources may influence the performance of diffusers.

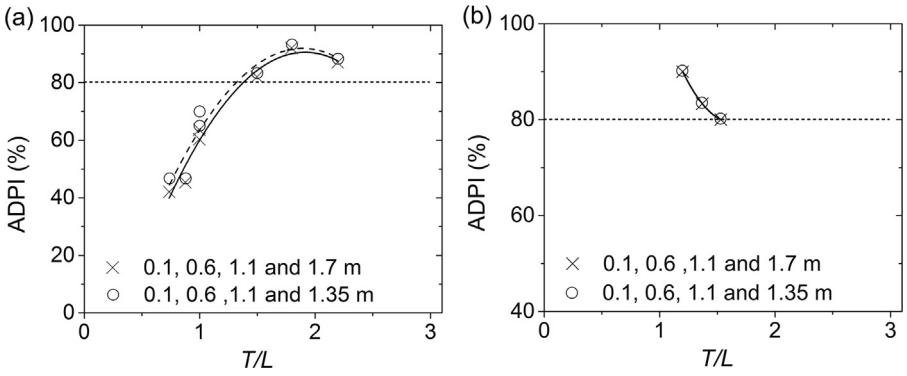


Fig. 5. The variation of ADPI using two sets of sampling locations for 50 W/m^2. (a) Adjustable blade grilles; (b) Nozzles. *T* is the throw length with a terminal speed of 0.25 m/s; *L* is room characteristic length.

Among the 15 diffuser types, ceiling diffusers generally perform better than grilles installed at high sidewalls or window sills, and better than diffusers at the ceiling periphery. Plaque face diffusers, ceiling square diffusers, linear slot diffusers, swirl diffusers, and louvered face diffusers can produce indoor air distribution with ADPI greater than 80% at all tested airflow rates. Affected the least by cooling load and airflow rate, those diffusers are most suitable for the application of variable air volume (VAV) systems.

Diffuser type is an important factor for indoor air distribution, comfort, and air quality. The minimum set-points of VAV systems depend on what types of diffusers are used. For most ceiling diffusers, with the exception of round ceiling and perforated diffusers, the minimum set-points could be less than 30% without sacrificing ADPI. High sidewall grilles often create air dumping and cause draft discomfort in the occupied zone but adjusting blades upwards may resolve this problem.

The new ADPI guideline will help in designing for proper indoor air distribution through terminal diffusers. The guideline can also guide the determination of minimum set-points for VAV systems.

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Appendix A.

A.1 Air diffusion performance index (ADPI) guide

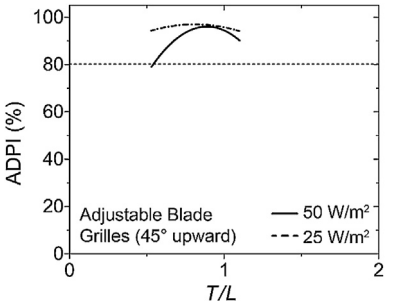
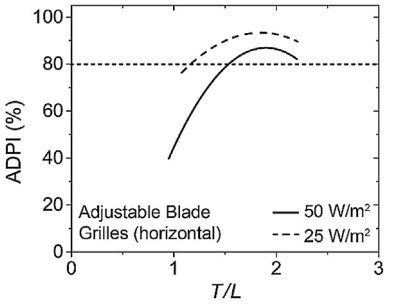
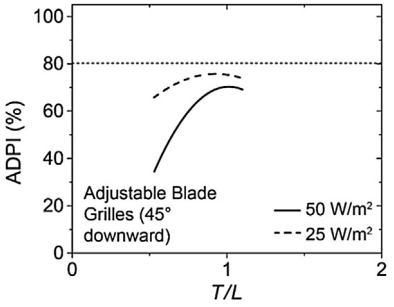
A.2 The effect of diffuser number on ADPI

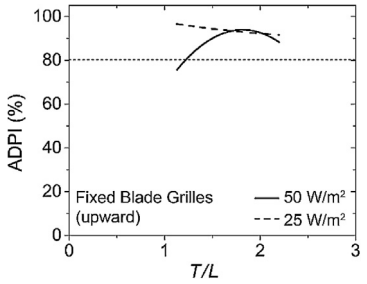
When the diffuser number in a room varies, given fixed total flow rate, the airflow rate and jet throw length of each diffuser are changed by the same factor as the diffuser number is. However, the change in magnitude might be different for the characteristic length of the room, which depends on how diffusers are spatially distributed. Consequently, the ratio of *T/L* and also ADPI may vary accordingly.

Fig. A1 presents one case of unchanged *T/L* and the other of changed *T/L* with a doubled diffuser number. The results show that both diffuser number and location influence ADPI. In other words, *T/L* and ADPI change when room characteristic length (*L*) varies by a different factor than diffuser number. The diffuser layout in Fig. A1(c) results in half the *T/L* ratio of those in the other two, which means that the variations of *T/L* and corresponding ADPI would be smaller for the case of Fig. A1(c). Hence, the configuration in Fig. A1(c) could have the best performance for VAV systems among the three.

For diffusers installed at a high sidewall, window sill or ceiling periphery, the room characteristic length does not change with diffuser number. As such, the condition of more diffusers for those types used in a room decreases *T/L* because airflow rate for each diffuser will be decreased. Nevertheless, when such diffusers are installed in multiple walls, window sills or ceiling peripheries, the interaction of multiple diffuser-jets might make indoor air distribution much more complex, which is not considered in this study.

Table A1
ADPI selection guide for overhead air distribution diffusers at low cooling loads.

Terminal device	Installation	Load (W/m^2)	*T/L* range (ADPI > 80%)	ADPI plot
Adjustable Blade Grilles	High sidewall (45° upward blades)	25 50	0.5–1.1 0.6–1.1	
	High sidewall (0° horizontal blades)	25 50	1.2–2.2 1.7–2.2	
	High sidewall (45° downward blades)	25 50	– –	

Fixed Blade Grills (High Sidewall installation)	High sidewall (15° upward blades)	25 50	1.1–2.2 1.2–2.2	

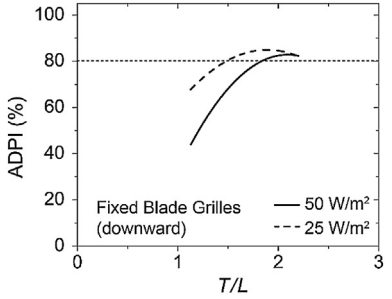
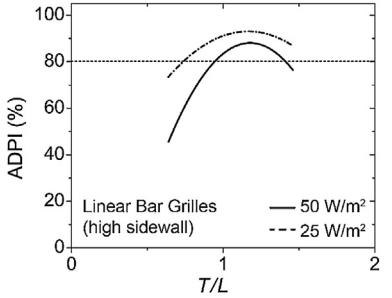
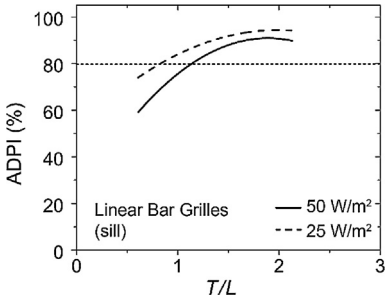
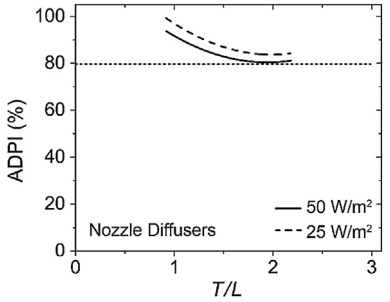
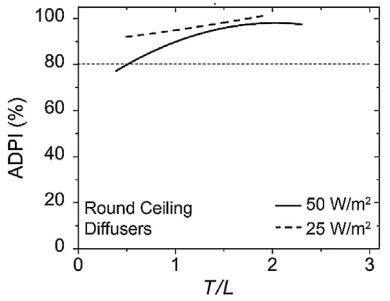
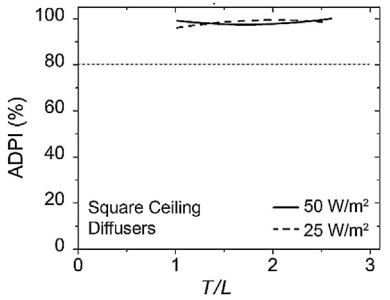
Terminal device	Installation	Load (W/m^2)	*T/L* range (ADPI > 80%)	ADPI plot
Fixed Blade Grills (High Sidewall installation)	High sidewall (15° downward blades)	25 50	1.6–2.2 1.9–2.2	
Linear Bar Grilles (High Sidewall installation)	High Sidewall	25 50	0.8–1.5 1–1.5	
	Sill	25 50	0.8–2.1 1.3–2.1	

Table A1 (*Continued*)

Terminal device	Installation	Load (W/m^2)	*T/L* range (ADPI > 80%)	ADPI plot
Nozzles (High Sidewall installation)	High sidewall	25 50	0.9–2.2 0.9–2.2	
Round Ceiling Diffuser	Ceiling center plane	25 50	0.5–2.3 0.7–2.3	
Square Ceiling Diffuser	Ceiling center plane	25 50	1–2.6 1–2.6	

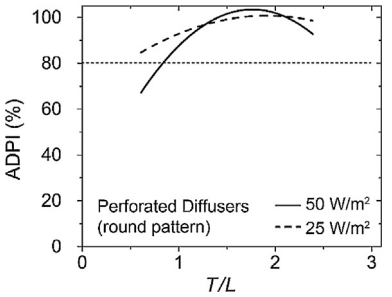
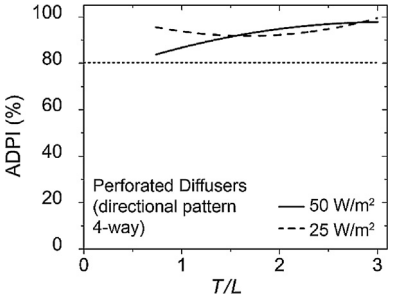
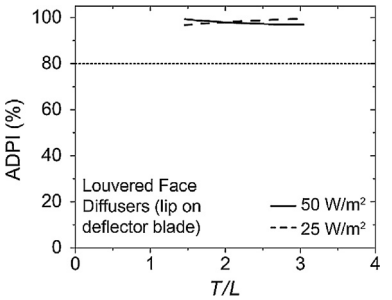
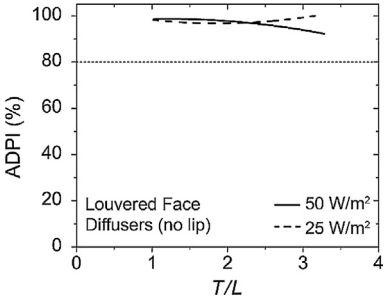
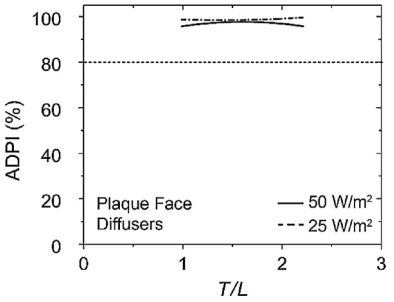
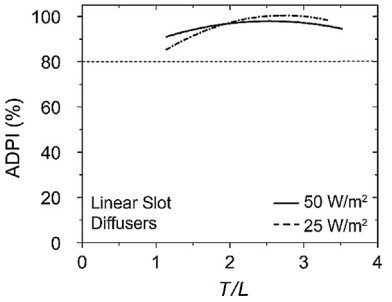
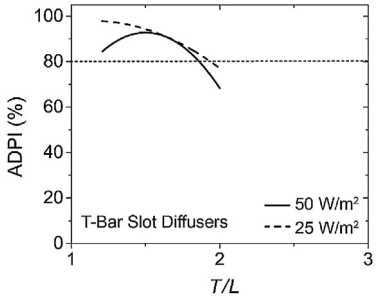
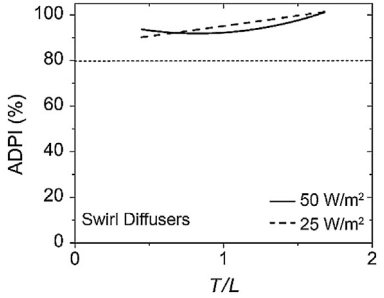
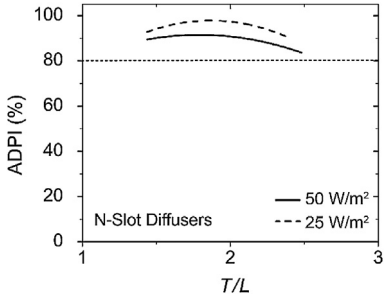
Perforated Diffusers-Round Pattern	Ceiling center plane	25 50	0.6–2.4 0.8–2.4	
Perforated Diffusers- Directional Pattern (4-way)	Ceiling center plane	25 50	0.7–3 0.7–3	
Louvered Face Diffusers-with Lip on Deflector Blade	Ceiling center plane	25 50	1.4–3 1.4–3	

Table A1 (*Continued*)

Terminal device	Installation	Load (W/m^2)	*T/L* range (ADPI > 80%)	ADPI plot
Louvered Face Diffusers-without Lip on Deflector Blade	Ceiling center plane	25 50	1–3.3 1–3.3	
Plaque Face Diffusers	Ceiling center plane	25 50	1–2.2 1–2.2	
Linear Slot Diffusers	Ceiling center plane	25 50	1.1–3.4 1.1–3.5	

T-Bar Slot Diffusers	Ceiling periphery	25 50	1.2–2 1.2–1.8	
Swirl Diffusers	Ceiling center plane	25 50	0.4–1.7 0.4–1.7	
N-Slot Diffusers	Ceiling periphery	25 50	1.4–2.4 1.4–2.4	

Note: "*T*" is the throw length of diffuser airflow with the terminal speed of 0.25 m/s, while "*L*" represents the characteristic length of a room.

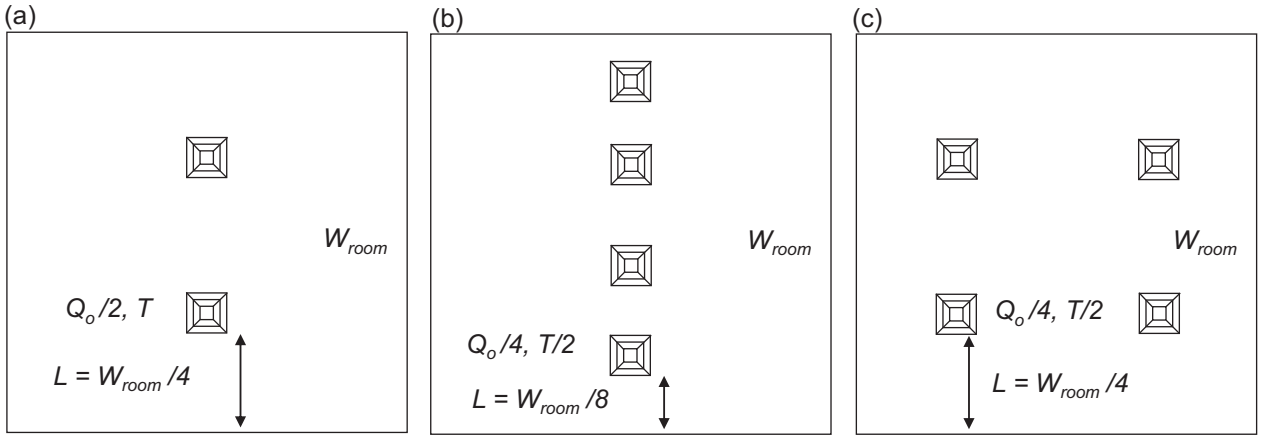


Fig. A1. The effect of diffuser number on T/L. (a) Original configuration with 2 ceiling diffusers, $T/L = 4\,T/W_{room}$ (room width); (b) Modified configuration with 4 ceiling diffusers in one row, unchanged $T/L = 4\,T/W_{room}$; (c) Modified configuration with 4 ceiling diffusers in two rows, halved $T/L = 2\,T/W_{room}$. T is the throw length with a terminal speed of 0.25 m/s; L is room characteristic length.

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